Fitting loss surfaces and diagnosing extrapolation

> **Reading note.** This is an explanatory reconstruction of Jiaxuan Zou's literature synthesis. Published experiments belong to the cited teams. Numbered protocols and illustrative matrices are our planning synthesis unless explicitly attributed.

The blog's §8 turns measured runs into a model of loss. First verify tuning completeness, warmup conventions, and compute accounting. A sophisticated regression cannot repair a run matrix that confounds size with horizon or changes recipes between points.

Candidate models

A baseline is the Chinchilla-style additive surface

$$\widehat{L}(N, D) = E + AN^{-\alpha} + BD^{-\beta},$$

with positive A, B, α, β and an estimated irreducible floor E. A coupled alternative discussed by the blog is

$$\widehat{L}(N, D) = E + \left(AN^{-\alpha} + BD^{-\beta}\right)^{k},$$

where $k > 0$ and $k = 1$ recovers the additive model. More parameters improve flexibility but can worsen identifiability. Compare models on development scales in the intended extrapolation direction, not solely on training error. For MoE, a fit using active parameters alone is conditional on fixed sparsity and expert structure.

Estimation protocol

1. Freeze a table of independent runs, endpoint conventions, trajectory clusters, and fit/development/holdout assignments.
2. Declare whether residuals use loss, log loss, or another transformation. State weights, bounds, and treatment of E.
3. Fit positive parameters in log coordinates using multiple initializations. Record the objective and converged solutions.
4. Inspect signed residuals against size, tokens, TPP, compute, and recipe. Coherent curvature suggests model mismatch.
5. Resample independent runs or shared-prefix clusters, refit the complete procedure, and recompute the engineering decision for every bootstrap replicate.
6. Compare predicted loss, allocation, and candidate ranking on untouched holdouts.

Under $C = 6ND$, minimizing the additive model gives

$$N_*(C) = \left(\frac{\alpha A}{\beta B}\right)^{1/(\alpha+\beta)} \left(\frac{C}{6}\right)^{\beta/(\alpha+\beta)}, \qquad D_* = \frac{C}{6N_*}.$$

The coupled model has the same minimizer under these assumptions because a positive outer power is monotone. This statement need not survive a change to architecture-dependent FLOPs or added constraints.

Trajectory fitting

When fitting intermediate losses, include schedule state and annealing explicitly. Many checkpoints on one trajectory are correlated measurements, not additional independent runs. A narrow interval obtained by treating them independently is misleading.

Deliverable: fit specification, residual plots, alternative-form comparison, bootstrap decision intervals, and a record of any failed extrapolation.

Sources: blog §8; Chinchilla, 2022; Skaling, 2026; Besiroglu et al., 2024; Tissue et al., 2024.

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